

QuantStudio HomeBrew Analysis Tool

User Manual — version 1.4.0

Spear Bio · Platform Development

September 22, 2026

What this tool does

It re-analyses a QuantStudio qPCR run with the **Spear Bio HomeBrew algorithm** instead of the instrument's own Design & Analysis (DA2) Cq. You give it the raw per-cycle fluorescence that DA2 exports; it normalises FAM against ROX, fits and removes each well's background, places a single plate-wide threshold so that your qPOS wells land on a target Ct, and reports a Cq for every selected well.

The result is a Cq that is consistent from plate to plate and anchored to your own controls, rather than to whatever threshold DA2 happened to choose.

Same algorithm as the desktop tool. This web app and *SPR Middleware v1.4* implement the same HomeBrew 1.4 algorithm with the same default parameters. Use whichever is convenient; the Cq should agree closely.

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1 Before you start: exporting from Design & Analysis

The app reads the **raw per-cycle detector data**, i.e. the **Raw Data** sheet, which holds the channel columns **X1_M1** (FAM channel) and **X4_M4** (ROX channel).

In DA2, open the run and use **Export**. Either of these works:

Export style	What to upload
Combined workbook	One .xlsx containing all sheets. Upload that single file.
Separate files	Upload the file whose name contains Raw Data . Either .xlsx or .csv is accepted.

You do not need the Results, Amplification Data or Multicomponent sheets — the app computes its own. Extra files are ignored as long as one of them is Raw Data.

2 Step-by-step

2.1 1 — Upload

Drag the export onto **Design and Analysis exported files**. On success the app prints either *Loaded combined workbook: ...* or *All key files loaded successfully!*

2.2 2 — Plate format

Choose **384-well (16×24)** or **96-well (8×12)**. This sets the grid and the default qPOS wells, so set it before selecting wells.

2.3 3 — Select the wells to analyse

Only selected wells are analysed, plotted and exported. Three ways to select, and they combine:

- **Bulk rules** — pick a *Row rule* (all / odd / even), a *Column rule* (all / odd / even), set *Mode* to Select or Deselect, then press **Apply full-plate selection**. Odd/even rules are the quick way to handle plates where only alternate columns are loaded.
- **Whole rows / columns** — choose from the two multi-selects, then **Apply row/col selection**.
- **The grid** — tick or untick individual wells directly.

The running count is shown as *N wells selected*, and the expander lists them.

Do not select empty wells. A well with no reaction mix has no real curve. The app guards against the obvious case — any well whose mean FAM *and* mean ROX are below 5% of the plate maximum is reported as no-Ct — but wells containing buffer only can still pass that test and add noise. Select what you actually loaded.

2.4 4 — qPOS (reference) wells

Pick the wells that carry the qPCR positive control. Defaults are **O24**, **P24** on a 384-well plate and **H6**, **H12** on a 96-well plate.

This is the single most important input. The threshold is chosen so the *average* Ct of these wells equals the target Ct, so every reported Cq on the plate is anchored to them. Pointing at the wrong wells shifts the whole plate.

2.5 5 — Parameters

Control	Default	Meaning
target ct	25.0	Ct the qPOS wells are driven to. Changes the threshold, and so every Cq on the plate.
background screening window	8	Width of the rolling window used to find where each curve lifts off the baseline.
shannon limit	50	Sensitivity of that liftoff detector; the step threshold is the curve's 5–95 % span divided by this number. Larger = more sensitive.
skipped cycles	5	Early cycles ignored before analysis starts.

Leave these at the defaults unless you have a specific reason. They are the validated HomeBrew 1.4 values and match SPR Middleware v1.4. Changing them changes reported Cq, and results produced with non-default settings are not comparable to anyone else's.

2.6 6 — Read the plot

The chart shows every selected well after ROX normalisation and background subtraction, on a **log** y-axis, with the fitted threshold as a dashed black line. Use it as a sanity check before trusting the numbers:

- curves should be flat early and rise smoothly;
- the threshold line should sit in the straight part of the rise, not up in the plateau nor down in the noise;
- a well that wanders below zero, or never crosses, will not produce a Cq.

2.7 7 — Export

Download QS-style Results (HomeBrew Cq) writes <run>_Results_SpearBio_HomeBrew.xlsx: a single **Results** sheet laid out like a DA2 export, with a metadata block and the table starting on row 26, so it drops into existing DA2-shaped workflows. Columns include the well, the HomeBrew Cq, and the baseline start/end actually used for that well.

The metadata block is filled with *placeholder* instrument fields (QuantStudio 5, 20 µL, and the current date/time) so the file matches the DA2 shape. Do not read those as a record of the real run — the run's identity comes from the file name.

3 How the number is produced

For each selected well, in order:

1. **Normalise** — divide the FAM channel (X1_M1) by the ROX channel (X4_M4), cancelling volume and optical differences.

2. **Find liftoff** — a rolling-window step detector locates the cycle where the curve leaves its baseline.
3. **Fit and remove background** — a robust (Theil–Sen) straight line is fitted to FAM and to ROX over the pre-liftoff region and subtracted. Being robust is the point: one off-trend early cycle cannot tilt the baseline.
4. **Set the threshold** — one threshold for the whole plate, solved so the mean qPOS Ct equals the target Ct.
5. **Interpolate Ct** — the crossing of the background-subtracted curve with that threshold.

A plate-level reference intercept (the median across wells, ignoring wells below 20% of maximum ROX) ties all wells to a common baseline.

4 Troubleshooting

Message or symptom	Cause and fix
No Raw Data file found	None of the uploaded files is a Raw Data export. Re-export from DA2 and include it.
Failed to read ...: Worksheet named 'Raw Data' not found	The uploaded workbook has no Raw Data sheet — e.g. a Multicomponent-only export. Upload the Raw Data one.
<i>Please select at least one qPOS well</i>	No qPOS chosen; nothing can anchor the threshold.
Threshold error, or no Cq anywhere	The qPOS wells probably do not amplify. Check you selected the right wells for <i>this</i> plate — qPOS positions differ between plate designs.
A few wells report no Cq	Either flagged empty (FAM and ROX both under 5% of plate max) or the curve never crosses the threshold.
Cq differs from DA2	Expected. DA2 uses its own threshold and baseline; HomeBrew anchors to your qPOS. Compare HomeBrew to HomeBrew.

5 Version

1.4.0 — HomeBrew 1.4: Theil–Sen robust baseline, defaults start cycle 5 / window 8 / Shannon limit 50 / target Ct 25.0. Verified against the SPR Middleware v1.4 reference on a 194-well plate: SD 0.013 cycles, maximum difference 0.14 cycles, correlation 0.999996.